Human Capital and Skill Formation

Labor I — Week 4

Teaching deck draft

Central question and motivation

- Week 3 asked how workers time labor supply given productive capacity.
- Week 4 asks where productive capacity comes from.
- Schooling, training, and childhood investments are all versions of the same dynamic tradeoff: pay now, produce later.

Why Week 3 is not enough

- Dynamic labor supply treats wages as moving objects, but it does not yet explain why wages rise over the lifecycle.
- Observed age-earnings profiles reflect accumulated skill, not only changing prices or preferences.
- We need an explicit state variable for productive capacity before we can interpret returns to skill.

Ben-Porath law of motion

$$h_{t+1} = (1 - \delta_h)h_t + A_t i_t^{\alpha}, \qquad w_t = r_t h_t$$

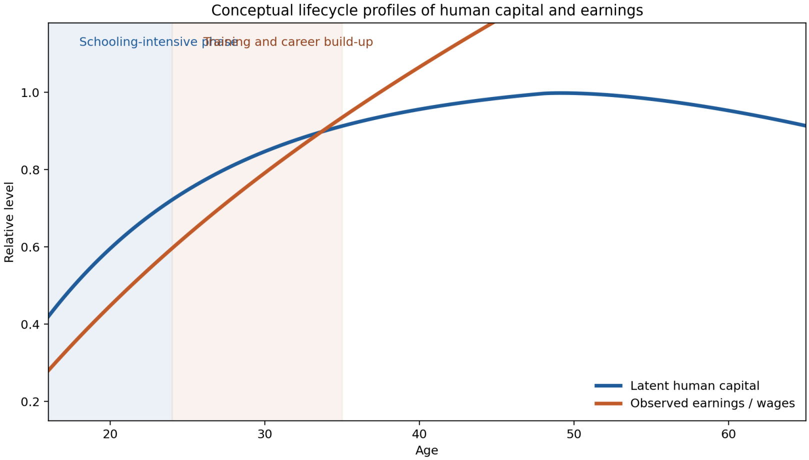
- Human capital is a state variable.
- Current investment raises future wages through future productivity.
- Remaining work horizon is what makes early investment especially valuable.

Investment tradeoff

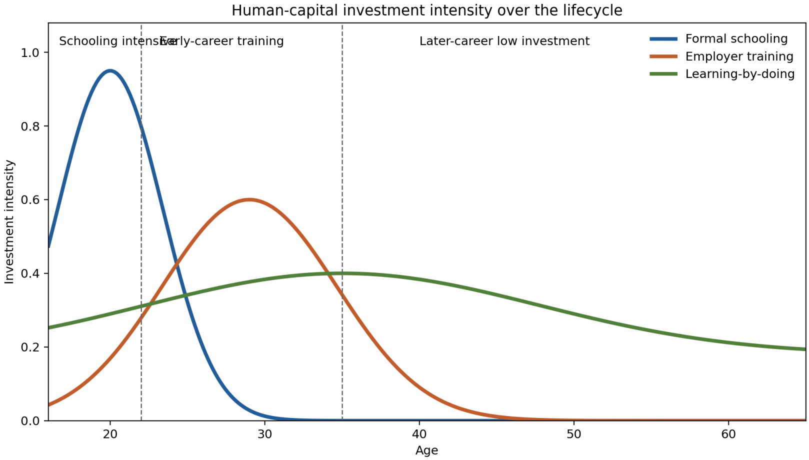
$$\mathsf{MC}_t(i_t) = \beta\, \mathbb{E}_t \left[\sum_{s=t+1}^{T} \beta^{s-t-1} \frac{\partial w_s}{\partial h_s} \frac{\partial h_s}{\partial i_t} \right]$$

- The cost side is tuition, effort, and foregone earnings.
- The benefit side is the discounted stream of higher future productivity.
- This is why lifecycle wage growth and schooling choice belong in one framework.

Lifecycle human capital and earnings



Schooling versus training as accumulation margins



General versus specific training

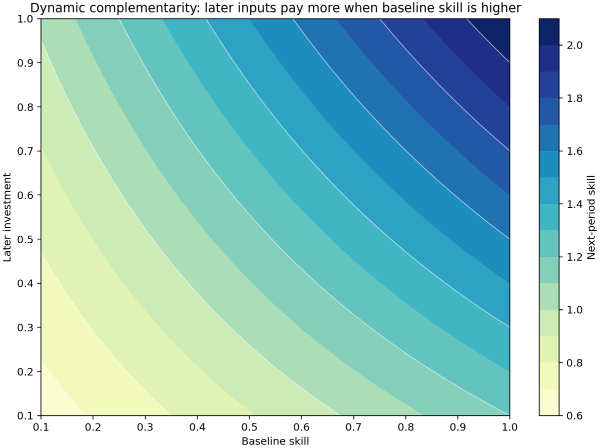
- Becker benchmark: firms should not pay for transferable general training in frictionless markets.
- Imperfect labor markets change the result.
- Wage compression, turnover frictions, and monopsony power can make firm-sponsored general training privately profitable.
- Training incidence is therefore informative, but not a complete measure of investment.

Self-productivity and dynamic complementarity

$$\theta_{t+1} = f_t(\theta_t, I_t, P_t)$$

- Self-productivity: current skill directly raises future skill.
- Dynamic complementarity: later investment is more productive when earlier skill is higher.
- Early environments matter because they change the payoff to later schooling and training.

Dynamic complementarity schematic



Empirical design map

Design	Main variation	Object learned
Schooling subsidy / draft-lottery	Exogenous schooling incentive	Reduced-form return on a narrow margin
Randomized childhood intervention	Program-induced input shift	Causal effect of bundled investments
Input-composition evidence	Center or input mix differences	Which inputs co-move with gains
Structural production function	Modeled skill dynamics	Technology parameters and complementarities
Credit-constraint design	Aid or borrowing environment	Whether finance distorts investment

Frictions, finance, and policy

- Borrowing constraints and limited commitment can distort high-return schooling investments.
- Information frictions change take-up and program choice even when subsidies are unchanged.
- Employer-side frictions shape who receives training and who captures the return.
- Human-capital policy therefore includes finance, advising, training support, and early-childhood inputs.

Core versus extension material

- Core lecture: Ben-Porath, schooling versus training, general versus specific training, self-productivity, design map, policy frictions.
- Extension block: deeper production-function estimation and schooling-finance models.
- Lab path: bounded synthetic reproduction in the spirit of Attanasio et al., then a quality-heterogeneity transfer in the spirit of Walters.

Bridge to Week 5

- Week 4 explains how skill is accumulated.
- Week 5 asks how labor markets price that skill.
- Observed wage returns mix human-capital accumulation, selection, and wage-setting.

Takeaways

- Human capital is a lifecycle state variable, not a fixed trait.
- Schooling, training, and childhood investment are different margins of the same dynamic problem.
- Early inputs matter partly because they raise the payoff to later inputs.
- Finance and information frictions can distort efficient investment and widen inequality.

Appendix: production-function extension

- Randomized interventions can identify treatment effects without identifying the full production technology.
- Structural estimation adds measurement and functional-form assumptions to recover self-productivity and complementarity.
- This is the natural extension slide if the class wants the full Attanasio et al. production-function discussion.

Appendix: schooling finance extension

- Credit frictions are not only low cash-on-hand.
- The key objects are collateral, limited commitment, repayment risk, and when returns arrive.
- This is the natural extension slide if the class wants the Lochner–Monge-Naranjo policy bridge.